

Birla Institute of Technology & Science, Pilani
Work Integrated Learning Programmes Division
First Semester 2026-2027

Mid-Semester Test
(EC-2 Regular)

Course No. : FST ZG619
Course Title : Finite Element Analysis
Nature of Exam : Closed Book
Weightage : 30%
Duration : 2 Hours
Date of Exam : 20/09/2026(AN)

No. of Pages	= 2
No. of Questions	= 3

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

- Q.1. For the bar shown in the Figure 1, determine the unknown displacement using FEA. Assume the constant Young's modulus E (kN/m^2) and area of cross section A (m^2) throughout. The linearly varying load has a magnitude zero at $x=1\text{m}$ and 8kN at $x=3\text{m}$. Determine the reactions at the fixed support. Draw the displacement, strain and stress profile over the length of the bar. [10 Marks]

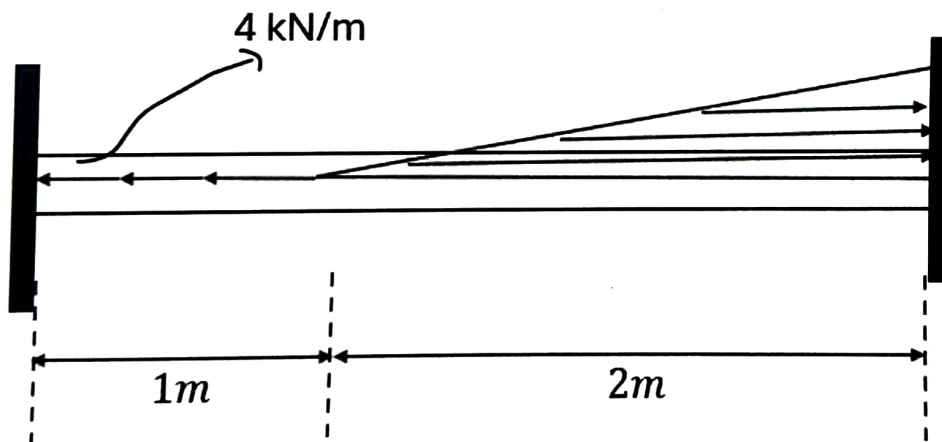


Figure 1

- Q.2. Analyze the structure shown in the Figure 2 using FEA considering the advantage of symmetry. Assume all the joints to be pin joints. Determine the reactions at the supports. Assume the area as ' A ' (m^2) and the Young's modulus as ' E ' (kN/m^2) for all the members. [10 Marks]

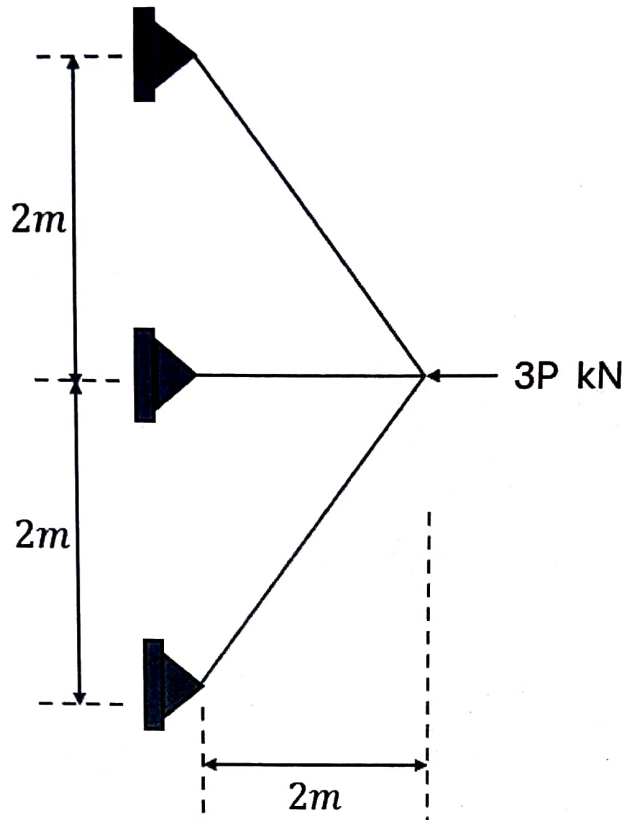


Figure 2

- Q.3. For the beam shown in the Figure 3, write the assembled stiffness matrix and the assembled force vector. Assume the Young's modulus as E (kN/m^2), moment of inertia as I (m^4) throughout the beam. [10 Marks]

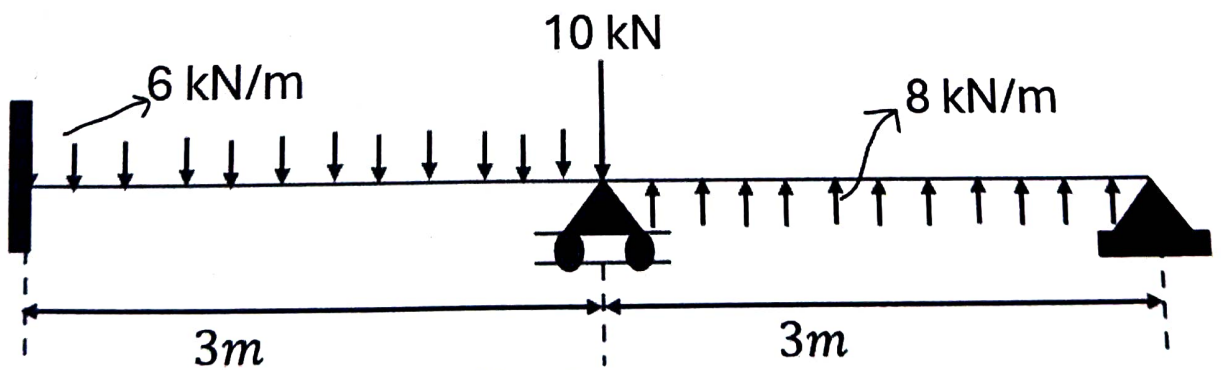


Figure 3

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Birla Institute of Technology & Science, Pilani
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First Semester 2026-2027

Mid-Semester Test
(EC-2 Regular)

Course No. : ST ZG551
 Course Title : Dynamics of Structures
 Nature of Exam : Closed Book
 Weightage : 30%
 Duration : 2 Hours
 Date of Exam : 20/09/2026 (FN)

No. of Pages	= 2
No. of Questions	= 5

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

- Q.1.** The multiple springs-mass system as shown in Figure 1 is subjected to free vibrations. The mass, $m = 24 \text{ kg}$, the stiffnesses of the springs are: $k_1 = 1000 \text{ N/m}$, $k_2 = 2000 \text{ N/m}$, $k_3 = 3000 \text{ N/m}$ and $k_4 = 4000 \text{ N/m}$. The initial displacement and initial velocity given to the mass are 0.05 m and 0.5 m/s respectively. If the displacement response is $u(t) = A \sin(\omega_n t + \phi)$, then neglecting damping, find the following:

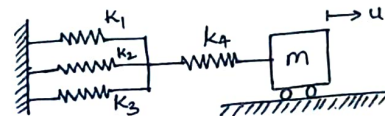


Figure 1

- (a) The effective stiffness of the system ..(1m)
- (b) The natural angular frequency ..(1m)
- (c) The amplitude of vibration ..(1m)
- (d) The phase angle, ϕ ..(1m)
- (e) The displacement at 1 seconds ..(1m)
- (f) The velocity when the mass passes through the equilibrium position ..(1m)

[6 Marks]

- Q.2.** The rigid frame shown in Figure 2 is fixed at the based and has a rigid top weighing 1000 kg . The columns are assumed to be massless and have lengths $L_1 = 4 \text{ m}$, $L_2 = 2 \text{ m}$. The flexural rigidity of the columns is taken to be $EI = 40 \times 10^6 \text{ Nm}^2$. The frame is subjected to free lateral vibrations, and damping is neglected.

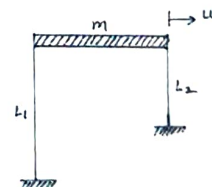


Figure 2

- a. Find the effective stiffness and time period. ..(2m)
- b. Write the expression for the free vibration of the system set by giving only an initial velocity of 2 m/s . ..(1m)
- c. Find the velocity and acceleration at 2 secs .. (1m)
- d. If the time period of the system has to be increased by 30 % what change must be made to the mass of the system. (2m)

[6 Marks]

- Q.3.** An empty elevated water tank is modeled as a SDOF system. A horizontal force of 300 kN causes a lateral displacement of 15 cm . The force is suddenly released, and the resulting free vibration is recorded. At the end of four complete cycles, the time is 2.0 sec and the amplitude is 10 cm . From these data compute the following: (a) damping ratio; (b) natural period of damped vibration; (c) effective stiffness; (d) effective weight; (e) damping coefficient; and (f) number of cycles required for the displacement amplitude to decrease to 5 cm .

[6 Marks]

- Q.4. A machine weighing 100 kg is mounted on a supporting system consisting of four springs and four dampers. The vertical deflection of the supporting system under the weight of the machine is measured as 2 cm. The dampers are designed to reduce the amplitude of vertical vibration to one-eighth of the initial amplitude after two complete cycles of free vibration. Find the following properties of the system: (a) undamped natural frequency, (b) damping ratio, and (c) damped natural frequency.

[6 Marks]

- Q.5. An air-conditioning unit with a mass of 320 kg is bolted at the middle of two parallel simply supported steel beams (Figure 3). The clear span of the beams is 3 m. The second moment of cross-sectional area of each beam is 450 cm^4 . The motor in the unit runs at 477.5 rpm and produces an unbalanced vertical force of 300 N at this speed. Neglect the mass of the beams and assume 1% viscous damping in the system; for steel $E = 200,000 \text{ MPa}$.

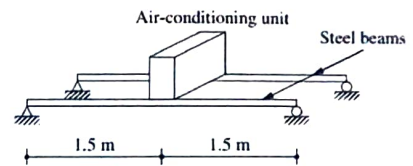


Figure 3

Determine the amplitudes of steady-state deflection and steady-state acceleration of the beams at their midpoints which result from the unbalanced force.

[6 Marks]

Birla Institute of Technology & Science, Pilani
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First Semester 2026-2027

Mid-Semester Test
(EC-2 Regular SET-II)

Course No. : ST ZG552
Course Title : Advanced Structural Mechanics and Stability
Nature of Exam : Closed Book
Weightage : 25% (As per Course Handout)
Duration : 2 Hours
Date of Exam : 19-09-2026 (AN)

No. of Pages	= 1
No. of Questions	= 4

Note to Students:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
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- Q.1. At a point in a three-dimensional body, the stress components with respect to the Cartesian coordinate system x, y, z are:

$$\sigma_x = 10 \text{ MPa}, \sigma_y = -6 \text{ MPa}, \sigma_z = 4 \text{ MPa}$$

and the shear stresses are:

$$\tau_{xy} = -4 \text{ MPa}, \tau_{yz} = 3 \text{ MPa}, \tau_{zx} = -5 \text{ MPa}$$

Determine a unit normal direction $\mathbf{n}$ such that the traction vector acting on a plane whose normal is $\mathbf{n}$ has $t_x = t_y = 0$.

[6 Marks]

- Q.2. For the state of stress specified in Q.1, construct the complete Cauchy stress tensor σ , and determine the principal stresses and principal directions.

[18 Marks]

- Q.3. For the state of stress given in Q.1., determine the first, second and third stress invariants

[6 Marks]

- Q.4. A three-dimensional stress state at a point is given by the following stress tensor in the Cartesian coordinate system x, y, z :

$$\sigma = \begin{bmatrix} 100 & 20 & 0 \\ 20 & 40 & 0 \\ 0 & 0 & 0 \end{bmatrix} \text{ MPa}$$

The coordinate system is rotated counterclockwise through an angle of 45° about the positive z -axis. Construct the transformation matrix and transformed stress tensor.

[20 Marks]

Mid-Semester Test
(EC-2 Regular Set -2)

Course No. : ST ZG612
Course Title : **ADVANCED STEEL STRUCTURES**
Nature of Exam : Closed Book
Weightage : 30% (100 Marks)
Duration : 2 Hours
Date of Exam : Saturday, 19/09/2026 (FN)

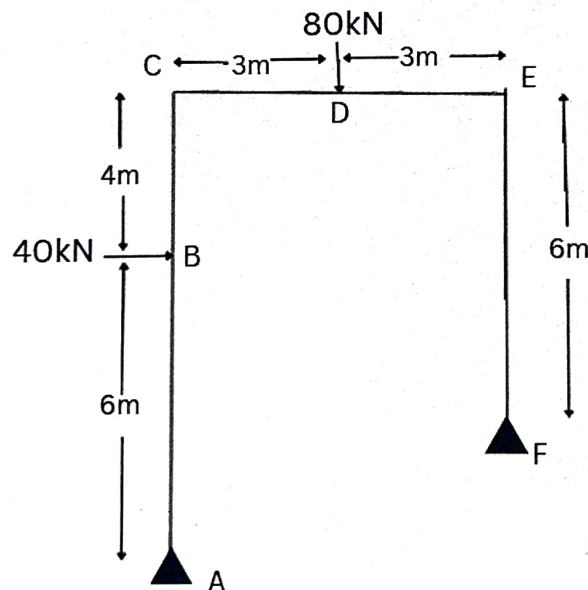
No. of Pages	=
No. of Questions	=

Note:

1. Please follow all the *Instructions to Candidates* given on the cover page of the answer book.
2. All parts of a question should be answered consecutively. Each answer should start from a fresh page.
3. Assumptions made if any, should be stated clearly at the beginning of your answer.

Note: Attempt all questions. Assume any missing data suitably. Use appropriate provisions of IS 800:2007 wherever required.

1. A I-section beam is fabricated with plates of following dimensions flanges 410mm x 20 mm and web 1400 mm x 15mm. Classify flanges, web and the section. Determine the bending capacity of the beam about its strong axis, Grade of steel Fe410. [30M]
2. A portal Frame ABCDEF of uniform cross section subjected to a 40kN horizontal load and a vertical load of 80kN is at D. Using principle of virtual work determine the plastic moment. [40M]



3. Determine the plastic section modulus about major axis and minor axis, the section dimensions $D = 225\text{mm}$, $B = 110\text{mm}$, thickness of flange 11.8mm and thickness of web $= 6.5\text{mm}$. (ISMB 225@ 306.07N/m) [30M]